Experiment 6

Aim :

To create a stored procedure transfer_employee(emp_id, new_dept_id) with validation and error handling, implement triggers for salary validation and audit logging, and test edge cases.

Source Code :

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-- EXPERIMENT 6
-- Stored Procedure, Validation, Triggers and Audit Logging


-- STEP 1: Create Audit Log Table
CREATE TABLE IF NOT EXISTS Salary_Audit (
    Audit_ID INT AUTO_INCREMENT PRIMARY KEY,
    Emp_ID INT,
    Old_Salary DECIMAL(10,2),
    New_Salary DECIMAL(10,2),
    Change_Date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    Action_Type VARCHAR(20)
);


-- STEP 2: Trigger for Salary Validation
DELIMITER //

CREATE TRIGGER validate_salary_insert
BEFORE INSERT ON Employee
FOR EACH ROW
BEGIN
    IF NEW.Salary IS NULL OR NEW.Salary <= 0 THEN
        SIGNAL SQLSTATE '45000'
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        SET MESSAGE_TEXT = 'Salary must be greater than zero';
    END IF;
END//

CREATE TRIGGER validate_salary_update
BEFORE UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF NEW.Salary IS NULL OR NEW.Salary <= 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be greater than zero';
    END IF;
END//

-- STEP 3: Trigger for Salary Audit Logging
CREATE TRIGGER audit_salary_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF OLD.Salary <> NEW.Salary THEN
        INSERT INTO Salary_Audit
        (Emp_ID, Old_Salary, New_Salary, Action_Type)
        VALUES
        (OLD.Emp_ID, OLD.Salary, NEW.Salary, 'UPDATE');
    END IF;
END//

-- STEP 4: Stored Procedure to Transfer Employee
CREATE PROCEDURE transfer_employee(
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    IN p_emp_id INT,
    IN p_new_dept_id INT
)
BEGIN
    DECLARE employee_count INT DEFAULT 0;
    DECLARE department_count INT DEFAULT 0;
    DECLARE current_dept_id INT DEFAULT NULL;

    DECLARE EXIT HANDLER FOR SQLEXCEPTION
    BEGIN
        RESIGNAL;
    END;

    -- Validate employee
    SELECT COUNT(*), MAX(Dept_ID)
    INTO employee_count, current_dept_id
    FROM Employee
    WHERE Emp_ID = p_emp_id;

    IF employee_count = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee does not exist';
    END IF;

    -- Validate department
    SELECT COUNT(*)
    INTO department_count
    FROM Department
    WHERE Dept_ID = p_new_dept_id;
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    IF department_count = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Department does not exist';
    END IF;

    -- Check whether transfer is necessary
    IF current_dept_id = p_new_dept_id THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee already belongs to this department';
    END IF;

    -- Transfer employee
    UPDATE Employee
    SET Dept_ID = p_new_dept_id
    WHERE Emp_ID = p_emp_id;

    SELECT 'Employee transferred successfully' AS Result;
END//

DELIMITER ;

-- STEP 5: Test Valid Employee Transfer
CALL transfer_employee(2, 2);

-- STEP 6: Check Employee Department
SELECT Emp_ID, Emp_Name, Dept_ID
FROM Employee
WHERE Emp_ID = 2;
```

-- STEP 7: Test Salary Update and Audit Trigger

UPDATE Employee

SET Salary = Salary + 1000

WHERE Emp_ID = 2;

-- STEP 8: Display Audit Log

SELECT * FROM Salary_Audit;

-- STEP 9: Test Edge Cases separately

-- Uncomment and run each CALL separately to see the error:

-- CALL transfer_employee(999, 2);

-- CALL transfer_employee(2, 999);

-- CALL transfer_employee(2, 2);

-- STEP 10: Test Invalid Salary separately

-- This should raise an error; do not run with the full script:

-- UPDATE Employee SET Salary = -500 WHERE Emp_ID = 2;

Output :

